Regression Discontinuity, RKD, Bunching, and Local Designs

Empirical Methods — Week 5

Teaching deck draft

Week 5 question

- What can be learned when rules, thresholds, kinks, dates, or borders create local variation?
- What exactly is discontinuous: treatment levels, treatment probabilities, incentive slopes, or densities?
- Why should potential outcomes be smooth at the threshold absent treatment?
- How local is the estimate, and is that local margin economically meaningful?

RD-style methods are design arguments before they are software commands.

Local designs as a family

Design	Identifying object	Core threat
Sharp RD	jump in outcome at cutoff	sorting or non-smooth potential outcomes
Fuzzy RD	local Wald ratio	weak or opaque first-stage jump
RKD	slope discontinuity ratio	noisy derivative estimation, bunching
Bunching	excess mass near kink or notch	model dependence, counterfactual density
Time RD	jump around a date	seasonality, serial correlation, anticipation
Spatial RD	jump across a border	geographic sorting, boundary confounds

Sharp vs fuzzy RD

$$D_i = \mathbf{1}\{X_i \geq c\}$$

$$\tau_{SRD} = \lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x]$$

$$\tau_{FRD} = \frac{\lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x]}{\lim_{x \downarrow c} \mathbb{E}[D_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[D_i \mid X_i = x]}$$

- Sharp RD: cutoff fully determines treatment.
- Fuzzy RD: cutoff shifts treatment probability; estimate is local to threshold compliers.

What RD identifies

- Identifies a local effect at the cutoff under continuity and no precise manipulation.
- Does not identify effects far from the cutoff without a portability argument.
- Does not identify mechanisms if crossing the threshold changes several channels.
- Does not survive sorting, heaping, or manipulation that changes composition at the cutoff.
- Strong applied interpretation names the threshold, local population, treatment margin, and unsupported claims.

Local polynomial estimation, bandwidths, and weights

Let $R_i = X_i - c$. A local linear RD solves a weighted least-squares problem:

$$\min_{\alpha_-,\alpha_+,\beta_-,\beta_+} \sum_i K\left(\frac{R_i}{h}\right) \mathbf{1}\{|R_i| \le h\} \left[Y_i - \alpha_- - \beta_- R_i - \mathbf{1}\{R_i \ge 0\} \left(\alpha_+ - \alpha_- + (\beta_+ - \beta_-)R_i\right)\right]^2$$

$$\widehat{\tau}_{RD} = \widehat{\alpha}_+ - \widehat{\alpha}_-$$

- Bandwidth h: larger means more precision and more bias risk.
- Triangular kernel: $K(u) = (1 - |u|)\mathbf{1}\{|u| \le 1\}$, emphasizing observations nearest the cutoff.
- Report bandwidth, kernel, polynomial order, and effective observations on each side.

`rdrobust` and `rdhonest`

`rdrobust` **logic**

- Estimate local-polynomial RD.
- Estimate leading bias with a higher-order fit.
- Re-center the estimate and adjust standard errors.
- Report conventional and robust bias-corrected inference.

`rdhonest` **logic**

- State a smoothness or curvature class.
- Bound worst-case bias inside the bandwidth.
- Build confidence intervals valid under that class.
- Often wider, but assumptions are explicit.

They address related but different inference questions.

Covariates and polynomial pitfalls

- Predetermined covariates can improve precision; they do not create RD identification.
- Covariates should be measured before treatment and be continuous at the cutoff.
- If covariates change the estimate sharply, treat that as a diagnostic warning.
- Avoid post-treatment controls, even if they improve fit.
- High-order global polynomials are discouraged because far-away data can shape boundary behavior.
- Prefer local linear or local quadratic fits with transparent bandwidth sensitivity.

Design diagnostics

Diagnostic	What it asks
Institutional rule	Is the threshold real, binding, and measured correctly?
Running-variable density	Is there sorting, heaping, or strategic timing near the cutoff?
Covariate continuity	Are predetermined characteristics smooth through the threshold?
Bandwidth sensitivity	Does the interpretation depend on one narrow window?
Placebo cutoffs	Are similar jumps appearing where no rule changes?
Inference level	Are mass points, clusters, time dependence, or spatial dependence handled?

Time RD vs event study

Time RD

- Running variable is calendar time.
- Cutoff is the treatment date.
- Identification is local continuity around the date.
- Best with sharp, unanticipated, isolated policy timing.

Event study / DID

- Uses comparison paths over time.
- Identification is untreated trend credibility.
- Best when untreated units or donor paths are strong.
- Handles broader dynamics when comparison paths are valid.

Time RD must confront seasonality, serial correlation, anticipation, and concurrent shocks.

Spatial RD

$$\tau_{border} = \lim_{d \downarrow 0} \mathbb{E}[Y_i \mid \text{signed distance}_i = d] - \lim_{d \uparrow 0} \mathbb{E}[Y_i \mid \text{signed distance}_i = d]$$

- One-dimensional version: signed distance to a border.
- Multidimensional version: local comparison around a boundary curve or segments.
- Credibility requires local geographic continuity and limited sorting.
- Implementation often needs boundary-segment fixed effects, local windows, and spatial dependence corrections.
- The estimate is local to the border, not automatically a regional or national ATE.

RKD and bunching

$$\tau_{RKD} = \frac{\lim_{x \downarrow c} \frac{d}{dx}\mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \frac{d}{dx}\mathbb{E}[Y_i \mid X_i = x]}{\lim_{x \downarrow c} \frac{d}{dx}\mathbb{E}[B_i \mid X_i = x] - \lim_{x \uparrow c} \frac{d}{dx}\mathbb{E}[B_i \mid X_i = x]}$$

$$\widehat{B} = \sum_{z \in \mathcal{B}} \left[f(z) - \widehat{f}_0(z) \right], \qquad \widehat{b} = \frac{\widehat{B}}{\widehat{f}_0(z^*)}$$

- RKD uses a slope change in incentives or treatment intensity.
- Bunching uses excess mass near a kink or notch plus an optimization model.
- Kinks change marginal incentives; notches create a level discontinuity and often a dominated region.

Choosing among local designs

Use when	Preferred design	Key caveat
Treatment jumps at cutoff	sharp RD	local external validity
Eligibility shifts take-up	fuzzy RD	first-stage content
Marginal incentive slope changes	RKD	derivative estimation and smoothness
Choices pile up at a kink/notch	bunching	behavioral model and counterfactual density
Policy begins on exact date	time RD	seasonality, anticipation, serial correlation
Institution changes at border	spatial RD	border confounds and sorting

Block 1 causal-inference summary

Method	Counterfactual logic	Main vulnerability
Selection on observables	conditionally similar units	unobserved selection, hidden spillovers
Experiments	random assignment	noncompliance, attrition, interference
DID / event study / SCM	untreated paths or donor pools	trend failures, anticipation, spillovers
IV / 2SLS	instrument-induced treatment variation	exclusion, weak first stage, local compliers
RD / RKD / bunching	local threshold or density comparison	sorting, locality, implementation sensitivity

Any credibility ranking is heuristic and context-dependent. Design fit beats method labels.

Research Lab design

Reproduce

- Close-election synthetic RD
- Local linear estimate
- Triangular weights
- RD plot

Diagnose

- Bandwidth sensitivity
- Covariate continuity
- Density checks
- RBC-style and honest intervals

Transfer

- Tax-kink bunching
- Counterfactual density
- Excess mass
- Elasticity memo

Takeaways

- RD-style designs are powerful because the comparison is local and institutional.
- Locality must be interpreted, not hidden.
- Bandwidths, weights, covariates, polynomial order, and inference define the effective comparison.
- Time and spatial RD need extra diagnostics because dates and borders carry their own confounds.
- Bunching is a density design plus a behavioral model.
- The best threshold papers make clear what the design identifies and what remains outside the threshold.